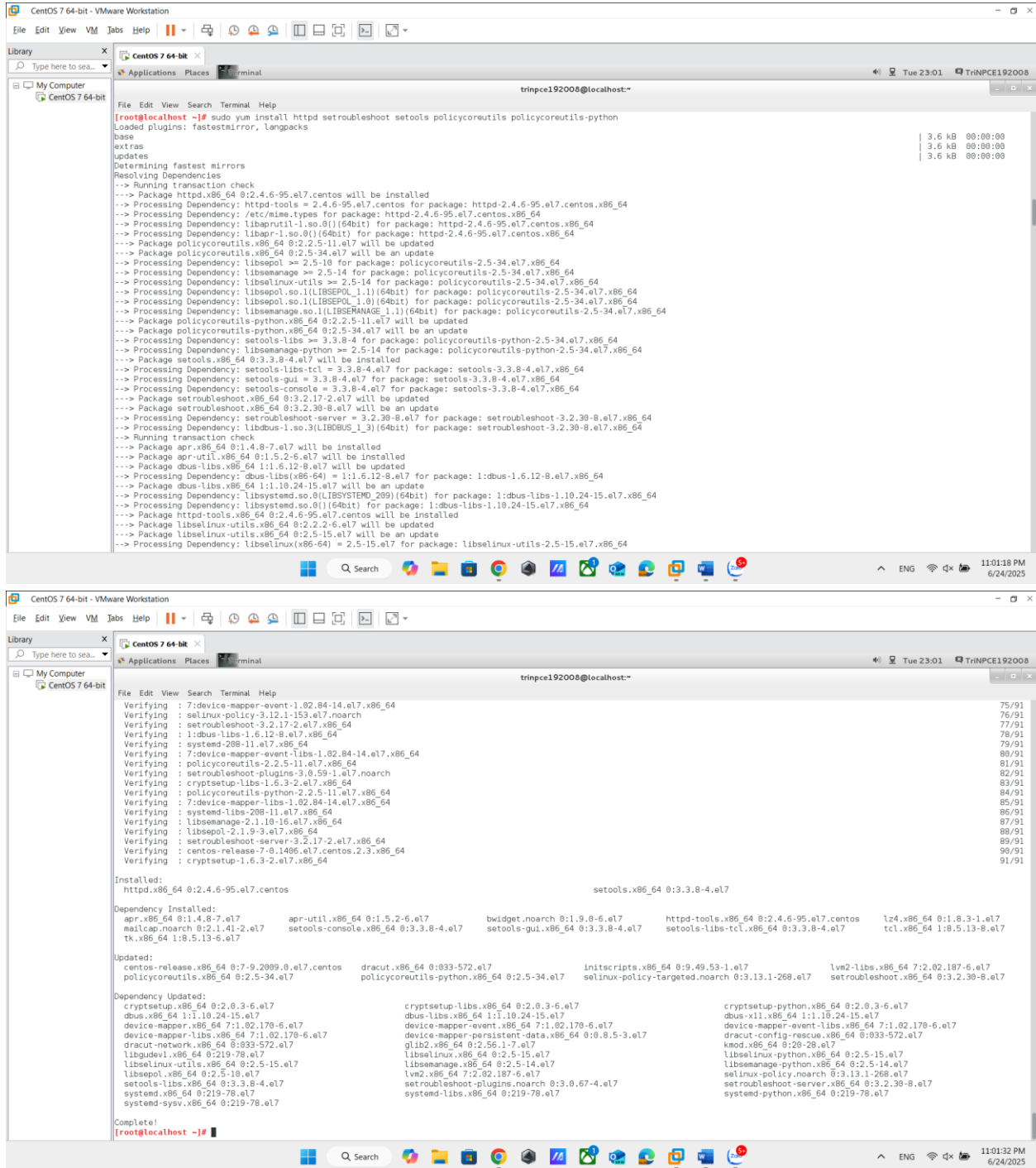
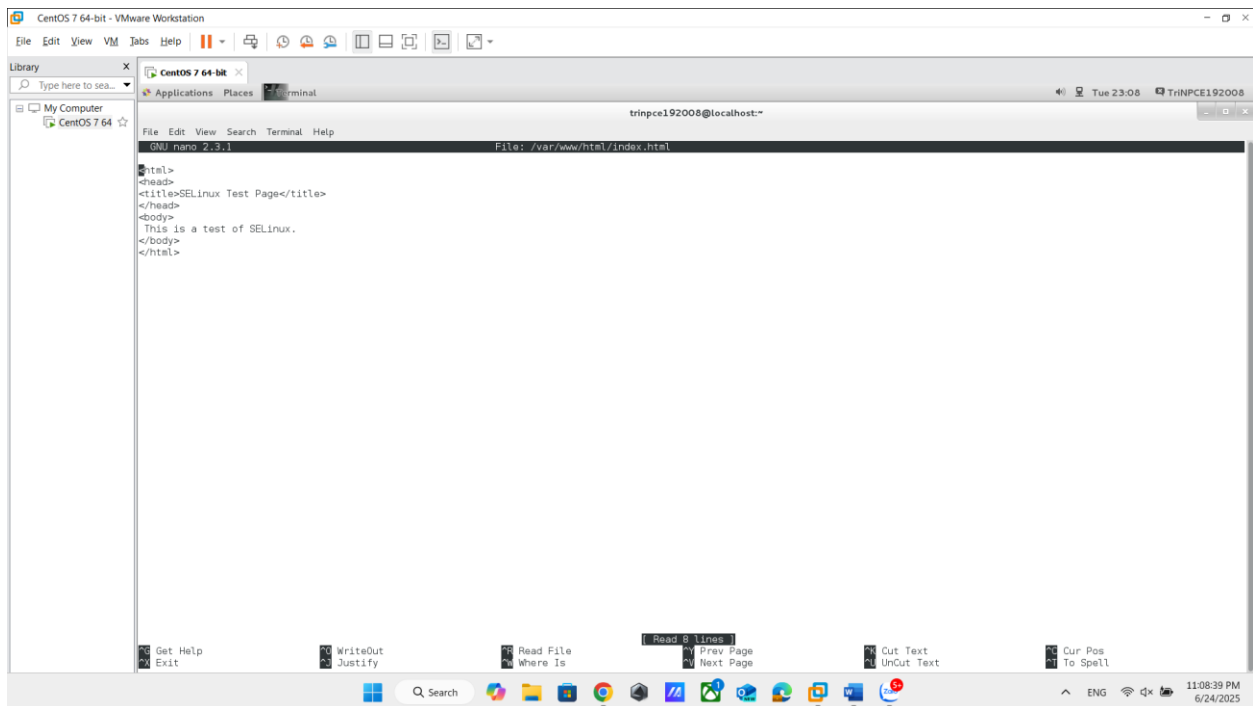
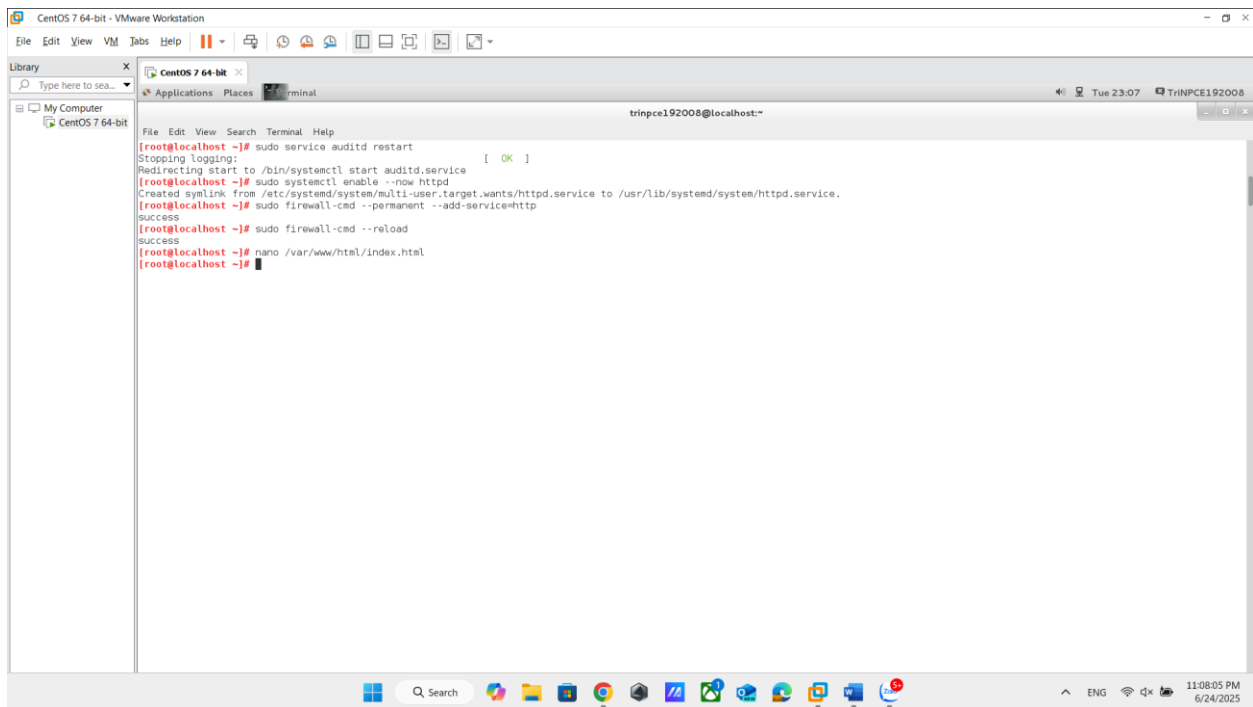


NGUYỄN PHẠM TRÍ LAB 6

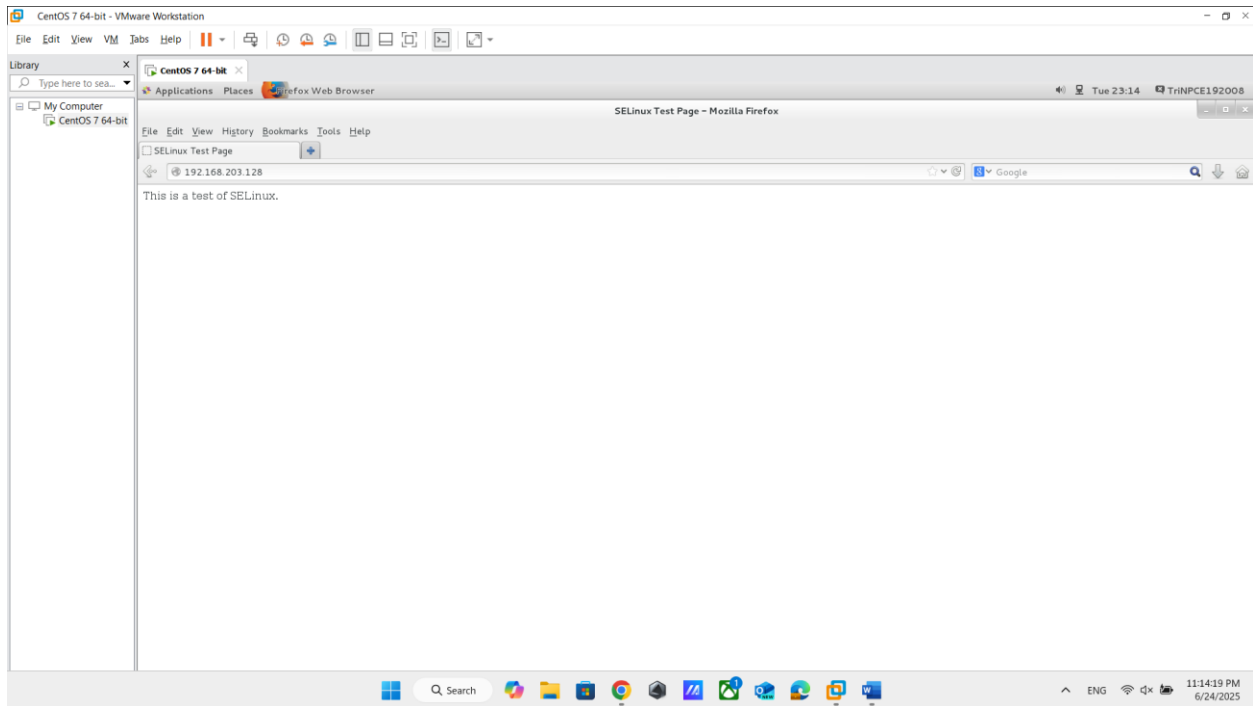
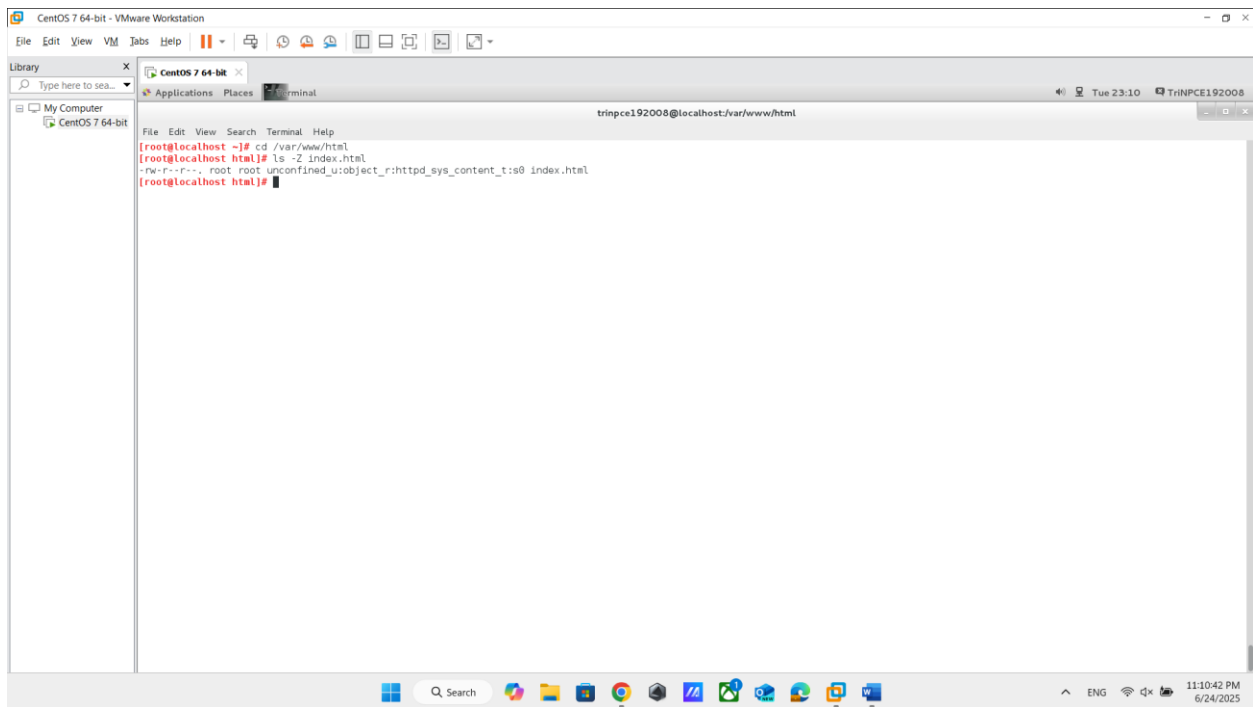
Hands-on lab – SELinux type enforcement



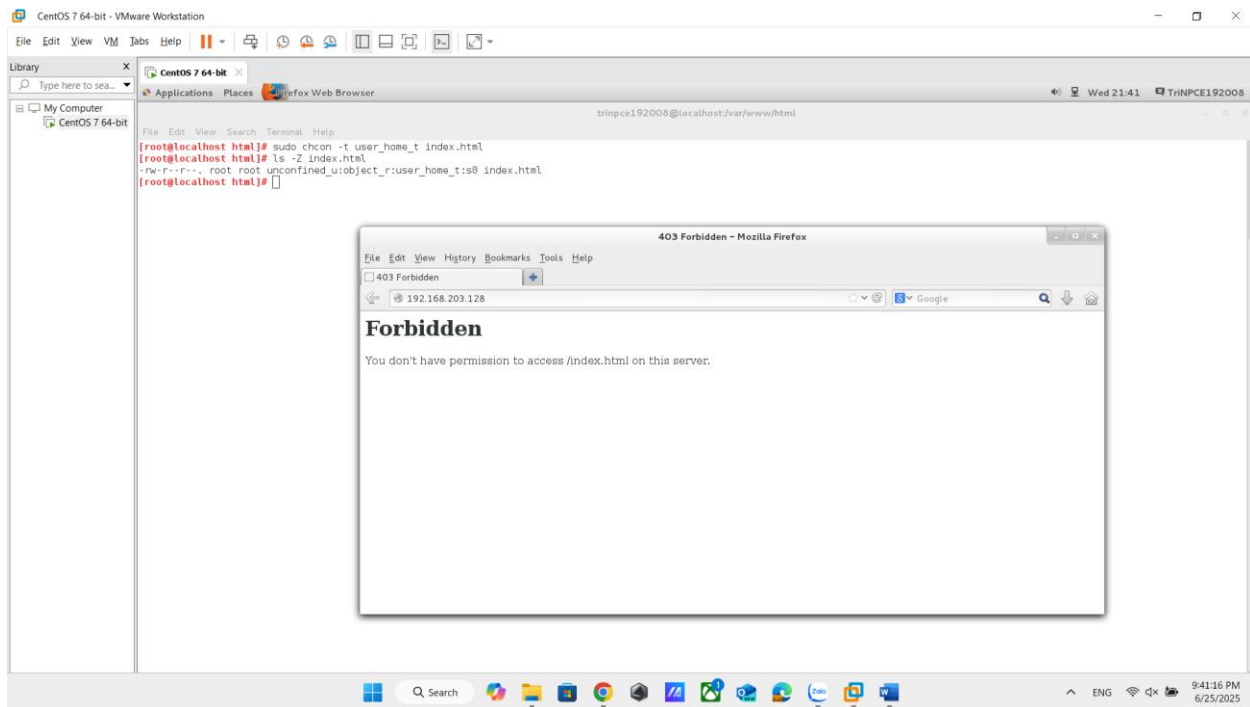
2 ảnh trên: Cài Apache cùng các công cụ SELinux



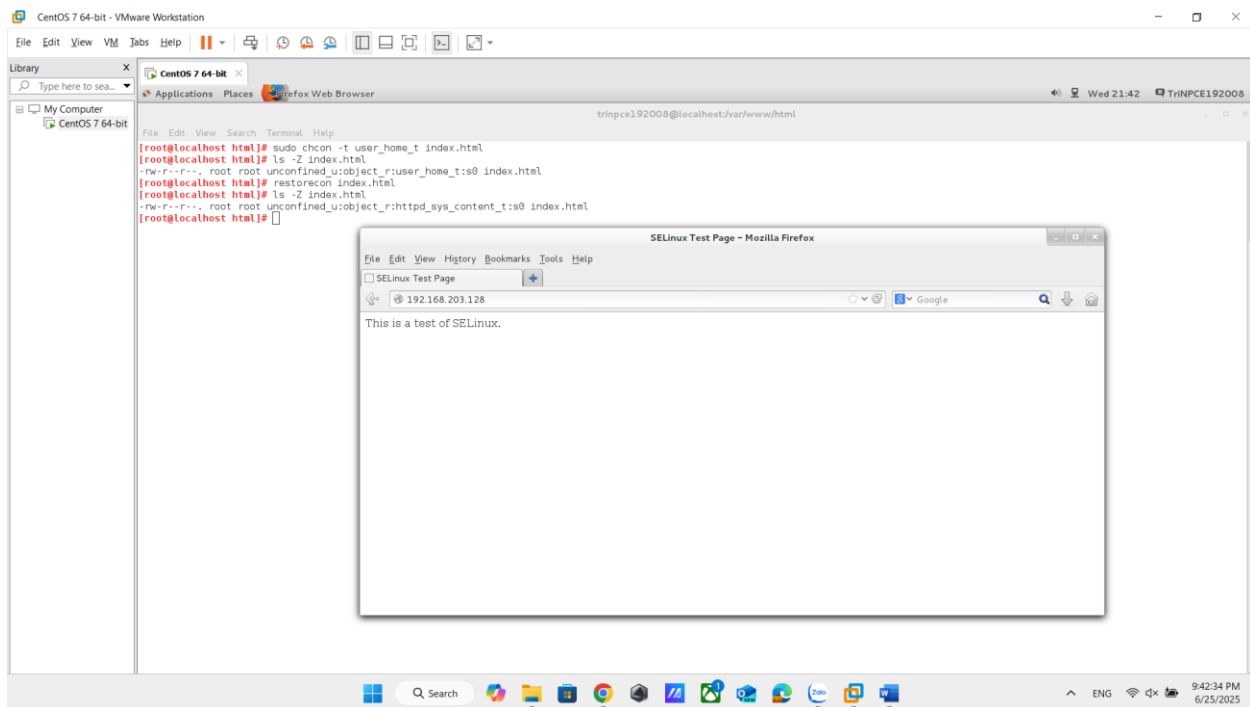
2 ảnh trên: Khởi động auditd để SELinux có thể log lại các hành vi bị từ chối, Mở cổng 80 và kích hoạt dịch vụ httpd, tạo tệp /var/www/html/index.html chứa mã HTML đơn giản



2 ảnh trên: Kiểm tra SELinux context và Truy cập trang từ trình duyệt để đảm bảo hoạt động bình thường.



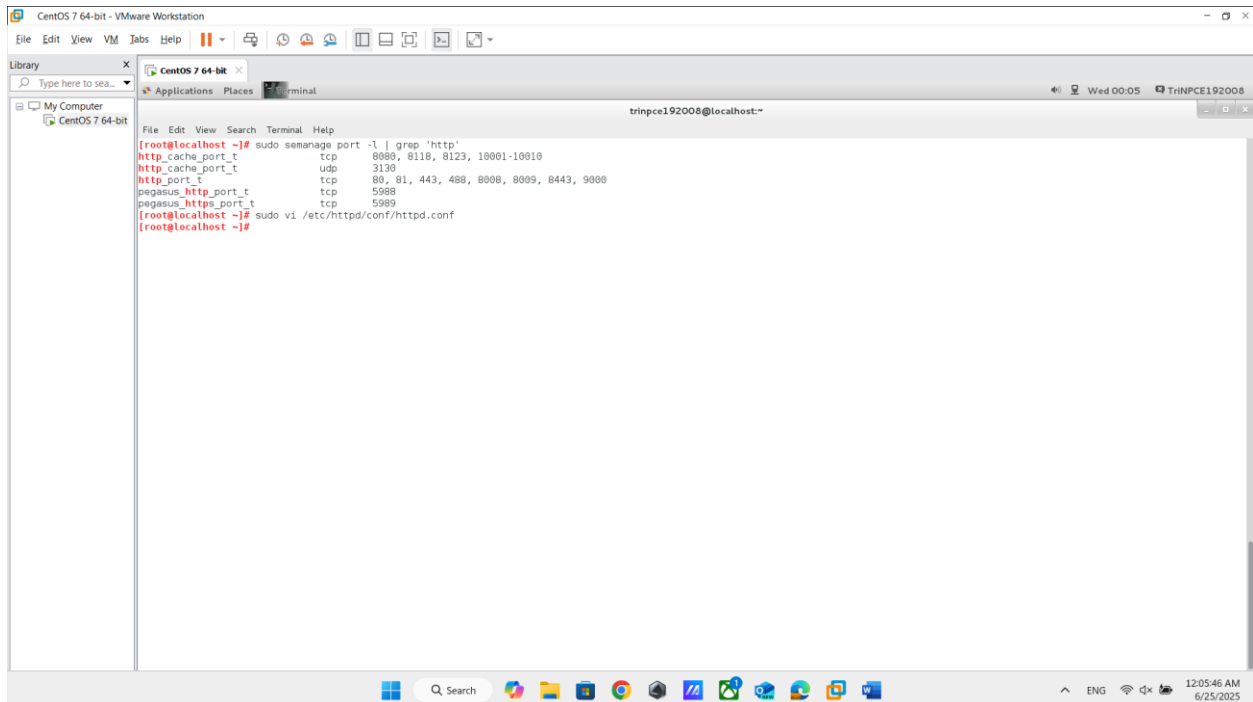
ảnh trên: Gây lỗi SELinux bằng cách đổi context, hay đổi **type** của file thành `user_home_t` không phù hợp với web content khiến Apache **bị chặn quyền truy cập**, và web sẽ bị lỗi



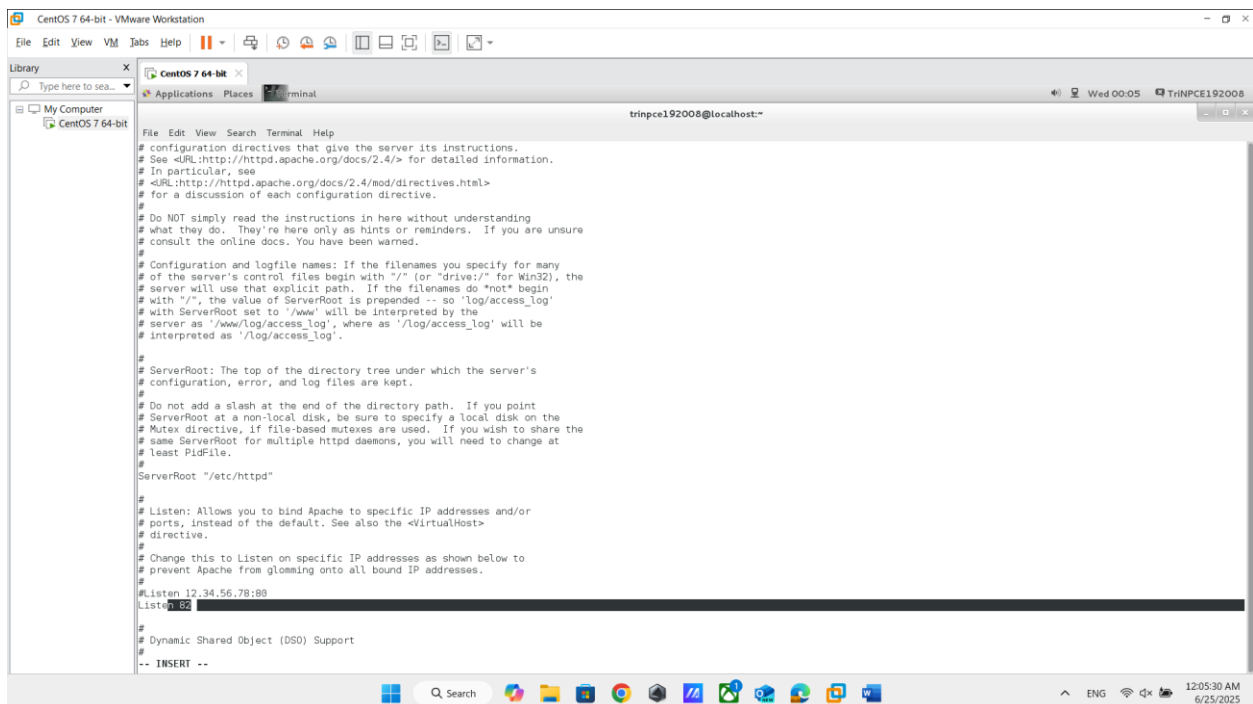
ảnh trên: Dùng `restorecon` đặt lại đúng context (`httpd_sys_content_t`), truy cập lại website → trang hoạt động bình thường trở lại

SELinux kiểm soát truy cập rất chặt chẽ: kể cả khi file nằm đúng thư mục, nếu context không đúng → dịch vụ sẽ bị chặn. Việc khôi phục context giúp nhanh chóng sửa lỗi mà không cần thay đổi cấu hình sâu.

Hands-on lab – SELinux Booleans and ports

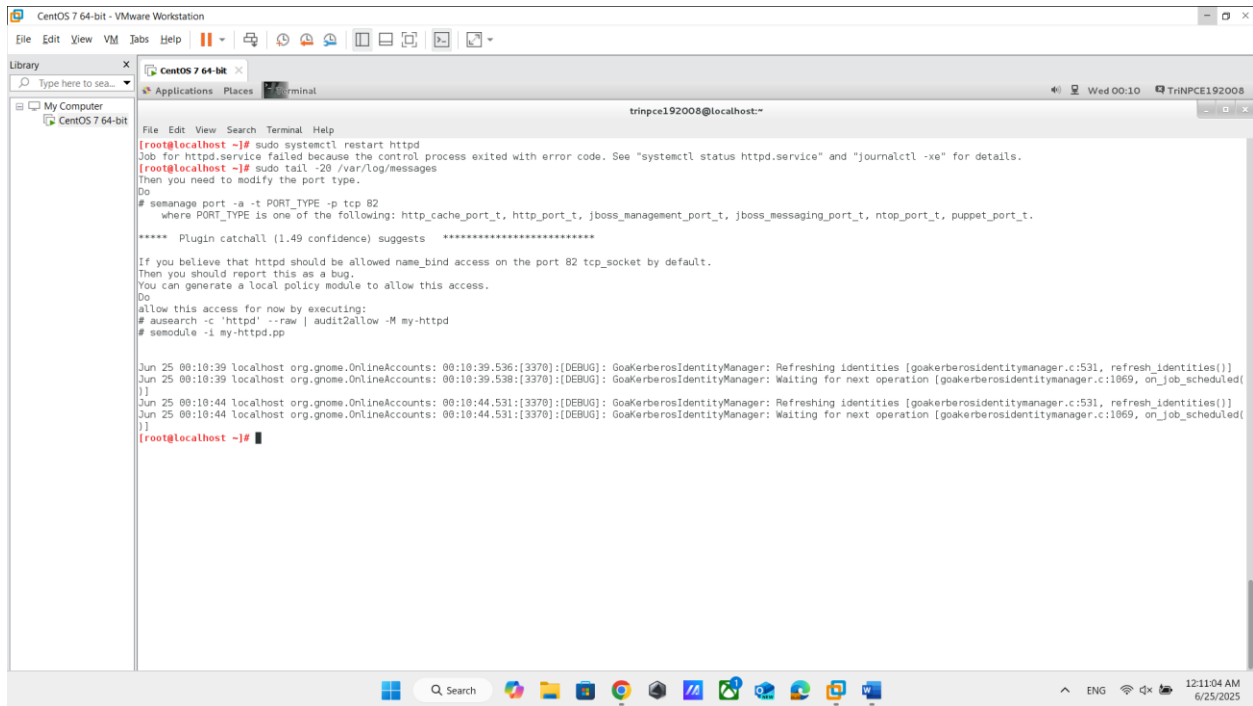


```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
CentOS 7 64-bit
trinpce192008@localhost:~$ sudo semanage port -l | grep 'http'
http_cache_port_t      tcp      8080, 8118, 8123, 10001-10010
http_cache_port_t      udp      3130
http_port_t            tcp      80, 81, 443, 488, 8080, 8089, 8443, 9000
pegasus_http_port_t    tcp      5988
pegasus_https_port_t   tcp      5989
trinpce192008@localhost ~$ sudo vi /etc/httpd/conf/httpd.conf
trinpce192008@localhost ~$
```



```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
CentOS 7 64-bit
trinpce192008@localhost:~$ cat /etc/httpd/conf/httpd.conf
# configuration directives that give the server its instructions.
# See <URL:http://httpd.apache.org/docs/2.4/> for detailed information.
# In particular, see
# <URL:http://httpd.apache.org/docs/2.4/ood/directives.html>
# for a discussion of each configuration directive.
#
# Do NOT simply read the instructions in here without understanding
# what they do. They're here only as hints or reminders. If you are unsure
# consult the online docs. You have been warned.
#
# Configuration and logfile names: If the filenames you specify for many
# of the server's control files begin with "/" (or "drive:/" for Win32), the
# server will use that explicit path. If the filenames do "not" begin
# with "/", the value of ServerRoot is prepended -- so 'log/access_log'
# with ServerRoot set to '/www' will be interpreted by the
# server as '/www/log/access_log', where as '/log/access_log' will be
# interpreted as '/log/access_log'.
#
# ServerRoot: The top of the directory tree under which the server's
# configuration, error, and log files are kept.
#
# Do not add a slash at the end of the directory path. If you point
# ServerRoot at a non-local disk, be sure to specify a local disk on the
# Mutex directive, if file-based mutexes are used. If you wish to share the
# same ServerRoot for multiple httpd daemons, you will need to change at
# least PidFile.
#
ServerRoot "/etc/httpd"
#
# Listen: Allows you to bind Apache to specific IP addresses and/or
# ports, instead of the default. See also the <VirtualHost>
# directive.
#
# Change this to Listen on specific IP addresses as shown below to
# prevent Apache from glomming onto all bound IP addresses.
#
Listen 12.34.56.78:80
Listen 80
#
# Dynamic Shared Object (DSO) Support
#
-- INSERT --
```

2 ảnh trên: liệt kê danh sách các cổng mà Apache (dùng context http_port_t) được phép lắng nghe, ta mở file cấu hình Apache đổi Listen 80 thành Listen 82



```
CentOS 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search...
My Computer
CentOS 7 64-bit
trinpce192008@localhost:~$ sudo systemctl restart httpd
Job for httpd.service failed because the control process exited with error code. See "systemctl status httpd.service" and "journalctl -xe" for details.
trinpce192008@localhost:~$ sudo tail -20 /var/log/messages
Then you need to modify the port type.
Do
# semanage port -a -t PORT_TYPE -p tcp 82
where PORT_TYPE is one of the following: http_cache_port_t, http_port_t, jboss_management_port_t, jboss_messaging_port_t, ntop_port_t, puppet_port_t.
***** Plugin catchall (1.49 confidence) suggests *****
If you believe that httpd should be allowed name_bind access on the port 82 tcp_socket by default.
Then you should report this as a bug.
You can generate a local policy module to allow this access.
Do
allow this access for now by executing:
# ausearch -c 'httpd' --raw | audit2allow -M my-httpd
# semodule -i my-httpd.pp

Jun 25 00:10:39 localhost org.gnome.OnlineAccounts: 00:10:39.536:[3370]:[DEBUG]: GoKerberosIdentityManager: Refreshing identities [goKerberosIdentityManager.c:531, refresh_identities()]
Jun 25 00:10:39 localhost org.gnome.OnlineAccounts: 00:10:39.538:[3370]:[DEBUG]: GoKerberosIdentityManager: Waiting for next operation [goKerberosIdentityManager.c:1869, on_job_scheduled()]
Jun 25 00:10:44 localhost org.gnome.OnlineAccounts: 00:10:44.531:[3370]:[DEBUG]: GoKerberosIdentityManager: Refreshing identities [goKerberosIdentityManager.c:531, refresh_identities()]
Jun 25 00:10:44 localhost org.gnome.OnlineAccounts: 00:10:44.531:[3370]:[DEBUG]: GoKerberosIdentityManager: Waiting for next operation [goKerberosIdentityManager.c:1869, on_job_scheduled()]
trinpce192008@localhost:~$
```

ảnh trên: Khởi động lại dịch vụ Apache và kiểm tra lỗi trong nhật ký hệ thống, ta thấy lỗi SELinux từ chối Apache sử dụng cổng 82 (vì chưa được cho phép qua semanage).

```
File Edit View Search Terminal Help
[root@localhost ~]# sudo tail -20 /var/log/messages
Then you need to modify the port type.
Do
# semanage port -a -t PORT_TYPE -p tcp 82
where PORT_TYPE is one of the following: http_cache_port_t, http_port_t, jboss_management_port_t, jboss_messaging_port_t, ntop_port_t, puppet_port_t.

***** Plugin catchall (1.49 confidence) suggests *****

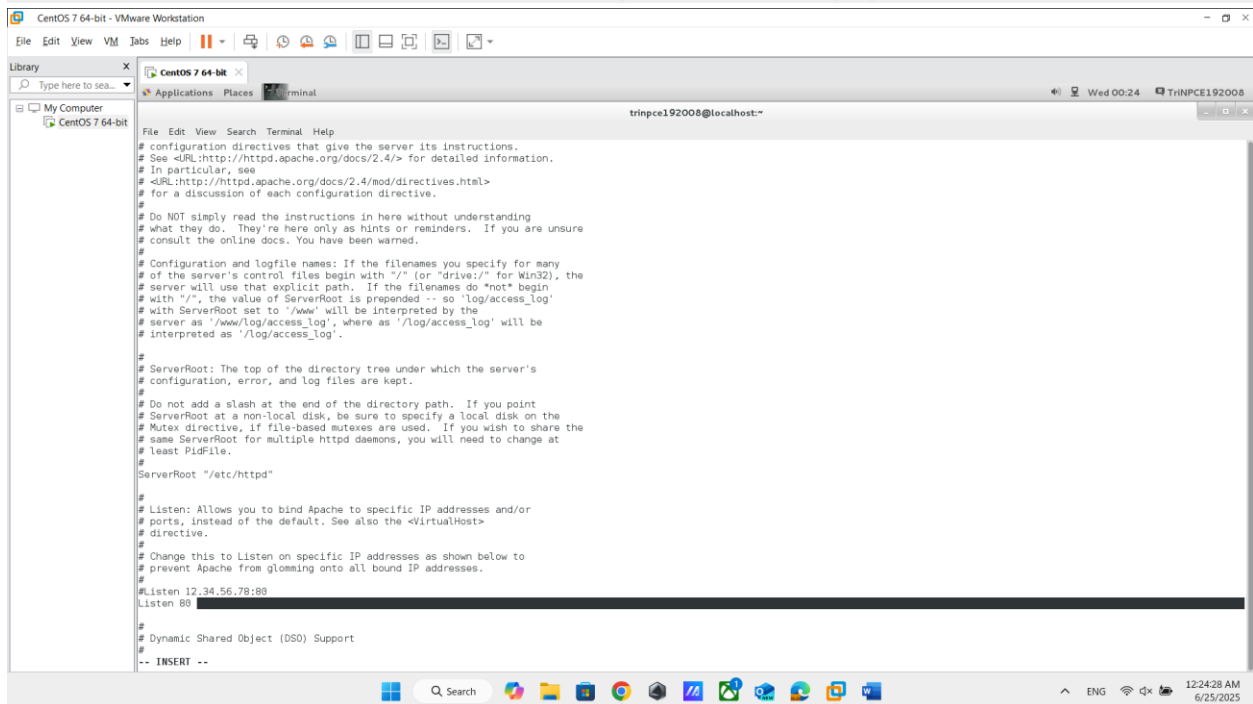
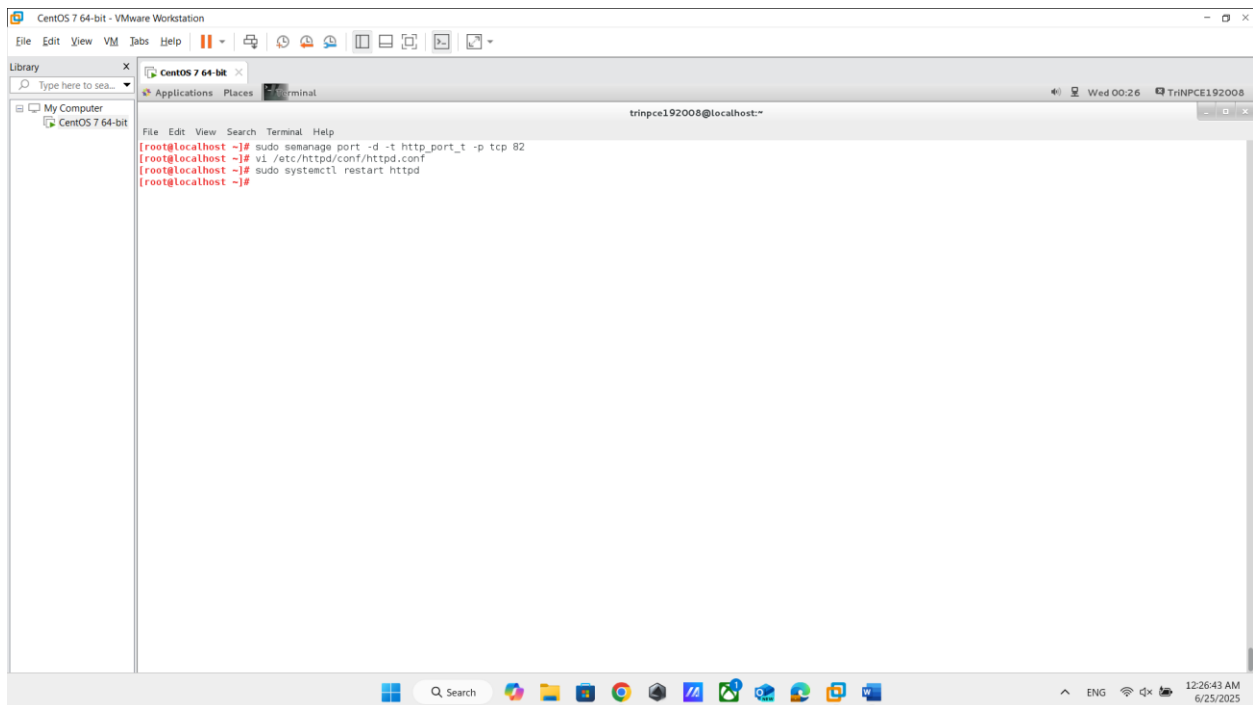
If you believe that httpd should be allowed name_bind access on the port 82 tcp_socket by default.
Then you should report this as a bug.
You can generate a local policy module to allow this access.
Do
allow this access for now by executing:
# ausearch -c 'httpd' --raw | audit2allow -M my-httpd
# semodule -i my-httpd.pp

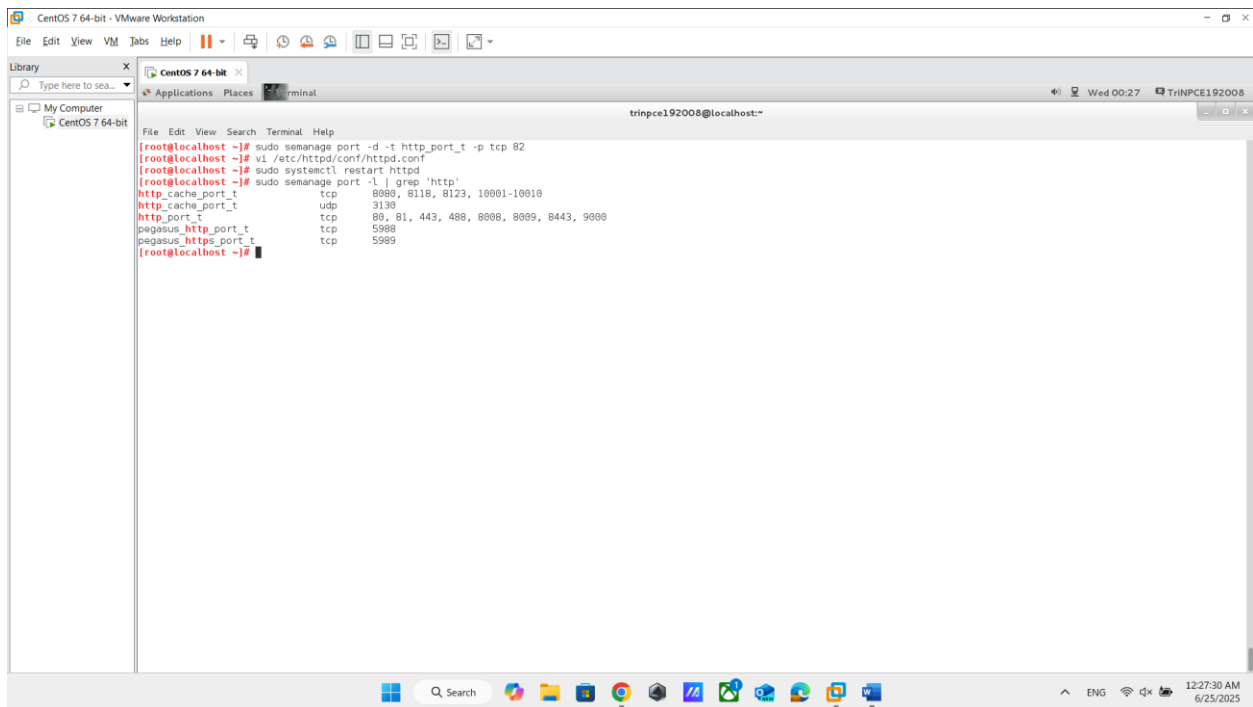
Jun 25 00:10:39 localhost org.gnome.OnlineAccounts: 00:10:39.536:[3370]:[DEBUG]: GoKerberosIdentityManager: Refreshing identities [goKerberosIdentityManager.c:531, refresh_identities()]
Jun 25 00:10:39 localhost org.gnome.OnlineAccounts: 00:10:39.538:[3370]:[DEBUG]: GoKerberosIdentityManager: Waiting for next operation [goKerberosIdentityManager.c:1069, on_job_scheduled()]
Jun 25 00:10:44 localhost org.gnome.OnlineAccounts: 00:10:44.531:[3370]:[DEBUG]: GoKerberosIdentityManager: Refreshing identities [goKerberosIdentityManager.c:531, refresh_identities()]
Jun 25 00:10:44 localhost org.gnome.OnlineAccounts: 00:10:44.531:[3370]:[DEBUG]: GoKerberosIdentityManager: Waiting for next operation [goKerberosIdentityManager.c:1069, on_job_scheduled()]

[root@localhost ~]# sudo semanage port -a 82 -t http_port_t -p tcp
[root@localhost ~]# sudo semanage port -l
SELinux Port Type          Proto  Port Number
-----
afs3_callback_port_t      tcp    7001
afs3_callback_port_t      udp    7001
afs_bos_port_t            udp    7007
afs_fs_port_t             tcp    2040
afs_fs_port_t             udp    7000, 7005
afs_ka_port_t             udp    7004
afs_pt_port_t             tcp    7002
afs_pt_port_t             udp    7002
afs_vl_port_t             udp    7003
agentx_port_t             tcp    705
agentx_port_t             udp    705
amanda_port_t             tcp    10080-10083
amanda_port_t             udp    10080-10082
amavisd_rcv_port_t        tcp    10024
amavisd_send_port_t       tcp    10025
amqp_port_t               tcp    15672, 5671-5672
amqp_port_t               udp    5671-5672
aol_port_t                tcp    5190-5193
```

```
File Edit View Search Terminal Help
[root@localhost ~]# sudo systemctl restart httpd
[root@localhost ~]# sudo semanage port -l | grep http
http_cache_port_t         tcp    8000, 8118, 8123, 10001-10010
http_cache_port_t         udp    3130
http_port_t               tcp    82, 80, 81, 443, 488, 8008, 8009, 8443, 9000
pegasus_http_port_t       tcp    5088
pegasus_https_port_t      tcp    5989
```

2 ảnh trên: Cấp quyền cho Apache (dùng SELinux type http_port_t) được phép lắng nghe trên TCP port 82, khởi động lại Apache và kiểm tra lại ta thấy cổng 82 đã được thêm vào





3 ảnh trên: xóa quyền sử dụng cổng 82 khỏi danh sách được phép của Apache và Vào lại file /etc/httpd/conf/httpd.conf, đổi lại Listen 82 thành Listen 80 khởi động lại apache. Cuối cùng kiểm tra lại ra thấy cổng 82 đã được xóa